DAEHO LEE

Researcher, Agency for Defense Development (ADD) | Daejeon, South Korea

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RESEARCH INTERESTS

Sensing and control at the foot–ground and human–robot interfaces of wearable and legged robots: gait-phase estimation, contact-aware load modulation, and human-intention recognition; perception-based control for unstructured environments.

EDUCATION

KAIST, Daejeon, South Korea Mar. 2020 – Aug. 2022
M.S. in Mechanical Engineering | GPA: 4.02/4.3
Thesis: *Terminal Decision Maker Against Upcoming Surface for the Safe Locomotion of Wearable Robots*
Research Advisor: Prof. Kyoungchul Kong

KAIST, Daejeon, South Korea Mar. 2013 – Feb. 2020
B.S. in Mechanical Engineering | Minor: Business and Technology Management | GPA: 3.85/4.3
Magna cum laude (military service 2016 – 2018)

RESEARCH EXPERIENCE

Exo Lab, Dept. of Mechanical Engineering, KAIST Mar. 2020 – Mar. 2023
Graduate Research Assistant; Researcher from Sep. 2022

- **Gait-phase sensing.** Developed a non-contact time-of-flight (ToF) sensing system with a threshold-based algorithm, enabling a lightweight, threshold-based gait phase classification with reduced computational overhead [J1, T1].
- **Initial-contact sensing and control.** Designed, built, and integrated a foot–ground distance module (custom PCB and ToF sensors) into Angel Legs M20 for initial-contact load-modulation control research [M1].
- **Force sensing and experiments.** Developed an FSR-based crutch-force module [J5] and a pneumatic plantar-pressure module; fabricated and instrumented modified WalkON Suit IV hardware and conducted the shank shock-absorption experiments [C1]. In the single-participant journal study, the absorber with the designed gait reduced mean peak heel ground-reaction force by 23% versus no absorber [J2].
- **Project leadership.** Led a six-month exoskeleton-dimension control project (KAIST Venture Research Program for Graduate Students, 2020): formed the team, wrote the proposal, and secured KRW 20 million. Built a data-collection system by adapting force-sensing hardware I had developed; a collaborating group performed the machine learning.
- **Experimental systems.** Integrated VICON, an AMTI treadmill, Delsys EMG, Xsens, and custom sensors for wearable-robot evaluation; developed MATLAB analysis software and OpenSim human–robot models.
- **Robot integration.** Assembled and debugged the team's CYBATHLON 2020 exoskeletons, including the pressure-sensing foot modules used for human-intention detection [J3].
- **Proposal development.** Co-wrote the human–wearable robot section of the selected NRF Leader Research Program proposal (PI: Prof. Kyoungchul Kong).

WORK EXPERIENCE

Agency for Defense Development (ADD) Apr. 2023 – Present
Researcher, 2nd R&D Institute – 2nd PMO | Daejeon, South Korea

- **IPS and standardization lead.** SECURITY CHECK PENDING (details restricted).
- **Software development.** SECURITY CHECK PENDING.
- **Project management.** SECURITY CHECK PENDING.

Journal Publications

[J1] **D. LEE**, "Real-time Gait Phase Detection System Using Non-contact Distance Sensors," *Journal of the Korean Society for Precision Engineering*, Vol. 43, No. 7, pp. 701–708, 2026. (in Korean)

[J2] J. Park, **D. LEE**, and K. Kong, "Shank Shock Absorption Mechanism and Associated Gait Pattern Design for Reduction of Ground Impact of a Powered Exoskeleton," *International Journal of Control, Automation and Systems (IJCAS)*, Vol. 21, No. 6, pp. 1959–1969, 2023.

[J3] J. Choi, K. Park, J. Park, **D. LEE**, *et al.*, "The History and Future of the WalkON Suit: A Powered Exoskeleton for People With Disabilities," *IEEE Industrial Electronics Magazine*, Vol. 16, No. 4, pp. 16–28, 2022.

[J4] E. Song, **D. LEE**, and K. Kong, "Design of Shoes Utilizing Torsion Spring for Increasing Gait Efficiency and Application in Wearable Robot," *Journal of Institute of Control, Robotics and Systems (ICROS)*, Vol. 27, No. 3, pp. 262–270, 2021. (in Korean)

[J5] K. Park, **D. LEE**, and K. Kong, "Development of FSR-Crutch Module for Real-Time Evaluation of Walking Performance in the Rehabilitative Applications," *Journal of Institute of Control, Robotics and Systems (ICROS)*, Vol. 26, No. 10, pp. 856–862, 2020. (in Korean)

Peer-reviewed Conference Proceedings

[C1] J. Park, **D. LEE**, K. Park, and K. Kong, "Reduction of Ground Impact of a Powered Exoskeleton by Shock Absorption Mechanism on the Shank," *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2085–2090, 2021.

Manuscripts in Preparation

[M1] **D. LEE**, J. Park, H. Choi, and K. Kong, "FDL: Foot–Ground Distance Layer for Initial-Contact Load Modulation in a Lower-Limb Exoskeleton," Manuscript in preparation.

Conference Presentations

[T1] **D. LEE**, K. Kong, and H. Choi, "Detection of Full Phases in a Gait Using Multi-Modal Sensor-Integrated Shoes," *The 30th ASME Annual Conference on Information Storage and Processing Systems (ISPS2021)*, Online, Jun. 2021.

Awards & Honors

Certificate of Commendation, Chief of Staff, *Republic of Korea Army*	2024
Global Leadership Award in "Challenge" Division, *KAIST*	2021
Gold & Bronze Medals (as team member), *CYBATHLON 2020 Powered Exoskeleton Race*	2020
KAIST Scholarship (full tuition), *KAIST*	2020 – 2021
National Excellence Scholarship for Science and Engineering (full tuition), *Ministry of Education (Korea)*	2015 – 2019
KAIST Scholarship (full tuition), *KAIST*	2013 – 2014

Technical Skills

Programming: Python, MATLAB, C/C++, LabVIEW

Embedded systems: STM32, sbRIO/FPGA, Arduino; PCB design, assembly, and debugging; SPI, I^2C, serial communications

Biomechanics: OpenSim, VICON, Xsens, AMTI, Delsys EMG

Mechanical design: SOLIDWORKS, AutoCAD, CATIA

Hardware: Robot fabrication and debugging (more than two years)

Languages: Korean (native), English (fluent)

Additional Experience

Samsung Electro-Mechanics — Engineering Intern Jul. 2019 – Aug. 2019

- Co-proposed a diagonal horizontally mounted capacitor (D-HMC) configuration to reduce multilayer ceramic capacitor (MLCC) acoustic noise. Suwon, South Korea.

AFN Korea, KATUSA (Mandatory Military Service) Aug. 2016 – Apr. 2018

Broadcaster and Video Editor; discharged as a Sergeant | Pyeongtaek, South Korea

- Filmed and edited feature stories and clips aired on AFN (American Forces Network) Korea.

Cairen (now WooriWel) — Undergraduate Intern Dec. 2015 – Jan. 2016

- Redesigned and modeled oxygen-respirator components in the CUop Program, Daejeon; continued the work in the capstone project "Oxygen Respirator for Elevator."